

LPIE Final Presentation Speaker Notes

Loan Performance Intelligence Engine (LPIE)

Intain FinTech AI Challenge 2026 - Final Round

Team: Narendra's Team | Sequence #9

Presentation Time: ~8 to 10 Minutes Total

SPEAKER PACING & DELIVERY GUIDELINES:

1. Speak in a calm, conversational, natural voice. Avoid rushing.
2. Pause for 1 second after stating key numbers (0.8168 AUC, 97.66 DQI, 38ms latency).
3. For Slide 11, point directly to the live JSON payload showing sub-50ms loan scoring.

SLIDE 1: Title & Introduction

Target Timing: 30 seconds

What to Say:

"Hello everyone and good afternoon judges. I am Narendra, presenting the Loan Performance Intelligence Engine - LPIE - for Narendra's Team, Sequence Number 9.

In simple terms, LPIE is an enterprise AI surveillance platform built for multi-vintage mortgage loan portfolios. It transforms raw monthly loan tapes into calibrated default hazard predictions, competing survival curves, and audit-ready credit notes in just 40.3 seconds.

Let's look at the industry problem we solve."

Transition: "Why does traditional secondary mortgage surveillance fail today?"

SLIDE 2: Bottlenecks vs. LPIE Paradigm Shift

Target Timing: 45 seconds

What to Say:

"Today, secondary mortgage surveillance has four major failure points:

1. Silent Delinquency Blindspots: Manual reviews miss 30 to 40 percent of early 30-to-60 day delinquencies. LPIE solves this with multi-horizon calibrated models that flag roll risk at 3 and 6 months early before default becomes irreversible.
2. Censoring Bias: Traditional credit models treat voluntary prepayment as non-default, creating positive survival bias. LPIE models default and prepayment together as competing risks via Cumulative Incidence Functions ($p = 2.83 \times 10^{-13}$).
3. Ledger Errors: Negative balances and impossible DPD jumps slip into securitization pools. LPIE catches these using 10 hard accounting checks synthesized with an unsupervised Isolation Forest.
4. Hallucination Bias: Generic AI chatbots invent credit narratives. LPIE uses a deterministic underwriting copilot where 100% of risk assertions cite exact numerical TreeSHAP attributions."

Transition: "Here is our end-to-end 9-phase architecture."

SLIDE 3: End-to-End System Architecture

Target Timing: 45 seconds

What to Say:

"LPIE is organized into 9 automated, sequential phases.

We ingest 34,285 monthly panel records across 3,000 loans spanning 2022 to 2024 vintages. Phase 1 scores data quality. Phase 2 engineers 12 domain features while quarantining 6 post-event administrative columns.

In Phase 3, we enforce a strict out-of-time chronological cutoff at November 2023 - training on historical vintages and validating on future periods with zero loan overlap.

This feeds our calibrated LightGBM and XGBoost models, competing-risks survival engine, anomaly scanner, TreeSHAP explainer, and grounded copilot.

The entire pipeline runs with a single command: `python run_pipeline.py` in just 40.3 seconds."

Transition: "Let's examine our data quality and drift audit."

SLIDE 4: Data Quality & Drift Audit

Target Timing: 40 seconds

What to Say:

"Before modeling, we verified our data integrity.

Our automated profiler gave this portfolio a Data Quality Index of 97.66 out of 100 - earning a certified Grade A rating. Over 91% of records passed all accounting checks with zero errors.

Completeness is 97.4%, using median imputation and missingness indicator flags to preserve statistical power.

And on the right, you can see our Population Stability Index audit. Every single demographic, origination, and credit feature scored a PSI below 0.04 - well below the Basel alert threshold of 0.10, proving zero out-of-time distribution drift."

Transition: "With clean data verified, let's look at predictive modeling."

SLIDE 5: Multi-Horizon Predictive Risk Modeling

Target Timing: 50 seconds

What to Say:

"For predictive risk modeling, our headline metric is an ROC-AUC of 0.8168 on 12-Month Default - a +12.6% improvement over our baseline model, with a PR-AUC of 0.5318.

For 12-Month Prepayment, we achieved 0.7119 ROC-AUC, a +23.8% lift.

Crucially, we calibrated our probabilities rather than just optimizing AUC. On the right, our reliability curves show our predicted probabilities hug the diagonal 45-degree line, giving us a low Brier score of 0.1063 - 43% lower error than baseline. When our model predicts a 15% default hazard, portfolio managers can trust that probability."

KEY PHRASE TO EMPHASIZE: "Out-of-time chronological validation with zero loan contamination."

Transition: "Now, why standard binary classification is not enough: competing risks."

SLIDE 6: Competing-Risks Survival & Hazard Engine

Target Timing: 50 seconds

What to Say:

"In mortgage portfolios, prepayment and default are mutually exclusive terminal states. A loan that prepays can never default. Ignoring this creates positive survival bias.

Across our 3,000-loan cohort, 12.6% defaulted, 55.7% prepaid early, and 31.7% remained active.

Our statistical discoveries:

- The Log-Rank test proves Prime versus Subprime survival curves diverge with $p = 2.83 \times 10^{-13}$.
- In our Cox Proportional Hazards model, Prime borrowers have a 26.9% lower monthly default hazard (HR = 0.731).
- Every +1% increase in interest rate accelerates default hazard by +57.8% (HR = 1.578).
- And our Weibull AFT model outperforms constant hazard by 149.8 AIC points, proving default hazard accelerates over loan seasoning."

KEY PHRASE TO EMPHASIZE: "Prepayment censors default - you must model them as competing risks."

Transition: "Next, how we catch anomalies and servicer errors."

SLIDE 7: Automated Anomaly & Exception Intelligence

Target Timing: 40 seconds

What to Say:

"Our anomaly engine combines 10 deterministic accounting rules with an unsupervised spatial Isolation Forest.

Here are four real case studies flagged by our system:

- Loan 2061: Current balance was higher than original loan amount - an obvious accounting error.
- Loan 676: DPD jumped from 0 to 90 in 30 days - skipping intermediate delinquency states.
- Loan 1925: Balance stayed completely frozen for 6 consecutive months with zero servicing updates.
- Loan 965: Servicer reported Current, but external credit bureau recorded 60 DPD.

For every case, the system automatically prescribes a servicer action, like issuing reconciliation notices or quarantining the tape."

Transition: "What happens under macroeconomic stress? Phase 5 simulations."

SLIDE 8: Macroeconomic Stress Testing & Scenario Lab

Target Timing: 45 seconds

What to Say:

"In Phase 5, we simulate what happens when macroeconomic shocks hit the portfolio.

Under Baseline, the 12-month default rate is 14.83% and prepayment is 50.75%.

Under Adverse Credit - simulating a +150 basis point rate shock with 7.5% unemployment - default rate surges to 33.91%, more than double.

At the same time, prepayment velocity drops by 40% to 30.53%.

This reveals a critical secondary market risk: when rates rise, defaults surge while portfolio duration extends because borrowers can no longer refinance. Our stress lab quantifies this tail risk instantly."

Transition: "How do we ensure full explainability and compliance?"

SLIDE 9: TreeSHAP Explainability & Fair Lending

Target Timing: 40 seconds

What to Say:

"Every prediction in LPIE is fully transparent and explainable using TreeSHAP.

Globally, the top three default drivers are days past due, balance-to-original ratio, and interest rate.

Locally, for any individual borrower, we decompose the prediction into exact plus-and-minus contribution values: $f(x)$ equals Base Value plus the sum of SHAP attributions.

This ensures strict compliance with ECOA and Regulation B Fair Lending standards - underwriters have the exact mathematical explanation ready for adverse action notices, with zero protected demographic variables ingested."

Transition: "Now, our Grounded Reviewer Copilot."

SLIDE 10: Grounded Reviewer Copilot (Zero Hallucination)

Target Timing: 45 seconds

What to Say:

"To help underwriters write credit memos faster, we built a Reviewer Copilot with a zero-hallucination guarantee.

It enforces three strict tiers:

1. Schema Grounding: It only accepts fields defined in our Data Dictionary.
2. Mandatory Citations: Every qualitative sentence MUST cite an exact numerical TreeSHAP attribution value - unsupported claims cannot be generated.
3. Immutable Audit Trail: 100% of prompts, outputs, and rejection events are logged to prompt_log.jsonl.

On the right, you can see an audit memo generated for Loan 421 citing exact SHAP values: 0 DPD reduced hazard by -0.184, and Prime credit reduced it by -0.112. It achieves a 0.00% hallucination risk score."

KEY PHRASE TO EMPHASIZE: "Zero hallucination - every risk claim cites a numerical TreeSHAP value."

Transition: "Let's examine our production microservice architecture."

SLIDE 11: Production Microservice Architecture & API

Target Timing: 40 seconds

What to Say:

"LPIE is engineered as a production-grade microservice suite.

On the left are our four architectural pillars: an asynchronous FastAPI REST engine, self-documenting OpenAPI 3.1 Swagger docs at /docs, Docker containerization, and an automated verification suite with 5 out of 5 Pytest suites passing at 100%.

And on the right side of the screen, you can see our live FastAPI microservice specification and the actual JSON payload. A client submits loan features - balance, rate, DPD, credit tier - and in just 38 milliseconds, the microservice returns 12-month default probability, prepayment probability, composite anomaly score, triage recommendation, and top SHAP attribution.

It handles over 2,500 requests per second with p99 latency under 45 milliseconds, ready for institutional trading desk integration."

KEY PHRASE TO EMPHASIZE: "Sub-50ms latency scoring microservice with OpenAPI 3.1 specifications."

Transition: "To conclude our presentation..."

SLIDE 12: Conclusion & Technical Discussion

Target Timing: 25 seconds

What to Say:

"To conclude: LPIE transforms secondary mortgage surveillance from slow, manual spreadsheets into an automated, explainable, and production-ready credit intelligence workflow - in 40.3 seconds.

Our full codebase, pipeline, and interactive REST API documentation are available on GitHub.

Thank you so much, judges. Narendra's Team is ready for your questions and technical discussion!"